Markdown - Simple Cheat Sheet

The basic syntax you need on your first day

Markdown for Beginners

2026

Start here

Markdown is **normal text with a few simple symbols for formatting**.

A Markdown file usually ends in:

`.md`

For example:

`notes.md`

You can write Markdown in **VS Code, GitHub, HackMD, StackEdit, Obsidian, Jupyter, or Quarto**.

> **Beginner rule:** type first, preview second. You do not need to memorize everything.

A good beginner workflow in VS Code is:

```
left: what you type        right: rendered preview
```

Open preview with:

```
macOS:          Shift + Command + V
Windows/Linux: Ctrl + Shift + V
```

---

1. Headings

Type this

```
# Main title
## Section
### Smaller section
```

It looks like this

Main title

Section

Smaller section

Simple rule: more `#` signs = smaller heading.

Always put a space after `#`.

```
## Results     good
##Results      avoid
```

---

2. Paragraphs

Type this

```
This is the first paragraph.

This is the second paragraph.
```

It looks like this

This is the first paragraph.

This is the second paragraph.

Simple rule: leave one blank line between paragraphs.

---

3. Bold and italic

Type this

```
**important result**
*paper title or emphasis*
***very important***
```

It looks like this

important result

paper title or emphasis

very important

A useful research-note pattern is:

```
**Finding:** The estimate is stable.

**Question:** Why does the result change?
```

```
**TODO:** Check Table 3.
```

---

4. Bullet and numbered lists

Bullets - type this

```
- Read the paper
- Check the appendix
- Reproduce Figure 2
```

It looks like this

- Read the paper
- Check the appendix
- Reproduce Figure 2

Numbered list - type this

```
1. Download the data
2. Clean the data
3. Run the analysis
```

It looks like this

1. Download the data
2. Clean the data
3. Run the analysis

---

5. Checkboxes

Checkboxes are useful for research tasks.

Type this

```
- [x] Read the introduction
- [ ] Check the data section
- [ ] Prepare seminar questions
```

It looks like this

- ☒ Read the introduction
- ☐ Check the data section
- ☐ Prepare seminar questions

  Checkboxes work especially well on GitHub and many Markdown editors.

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6. Links

The pattern is:

```
[text people see](web address)
```

Type this

```
[Google Scholar](https://scholar.google.com/)
```

It looks like this

Google Scholar

You can also link to another file in the same project:

```
[Open my notes](notes.md)
```

Simple rule: square brackets contain the label; parentheses contain the destination.

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7. Images

An image looks like a link with `!` at the beginning.

```
![description](path-to-image)
```

Suppose your folder is:

```
project/
  notes.md
  figures/
    result.png
```

Type:

```
![Main result](figures/result.png)
```

Simple rule: use a path relative to the Markdown file. Avoid paths such as `/Users/name/Desktop/...`.

If an image does not appear, check the filename and folder path first.

---

8. Inline code

Use one backtick when you want to show a filename, variable name, command, or code literally.

Type this

```
Open `analysis.R` and check the variable `income`.
```

It looks like this

Open `analysis.R` and check the variable `income`.

This is also useful for showing a LaTeX command such as `\qquad` without running it.

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9. Code blocks

Use three backticks before and after a larger block of code.

Type this

````
```r
model <- lm(y ~ x, data = df)
summary(model)
```
````

It looks like this

```r
model <- lm(y ~ x, data = df)
summary(model)
```

You may replace `r` with `python`, `stata`, `bash`, or another language name.

Simple rule: every opening set of three backticks needs a closing set.

---

10. A small table

Tables are optional. Learn them only when you need them.

Type this

```
| Variable | Mean |
|---|---:|
| Income | 52.4 |
| Age | 34.1 |
```

It looks like this

Variable	Mean
Income	52.4
Age	34.1

The second row tells Markdown that this is a table.

For complicated publication tables, use LaTeX, Quarto, or generated output instead of forcing everything into Markdown.

---

11. Simple math - optional

Some Markdown tools support LaTeX-style math (the notation many papers use for equations; you can just copy the examples).

Inline math

Type:

```
The parameter is $\beta$.
```

Rendered idea:

The parameter is β.

Display math

Type:

```
$$
y_i = \alpha + \beta x_i + \varepsilon_i
$$
```

Rendered idea:

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

Inside the math, `\alpha`, `\beta`, and `\varepsilon` are LaTeX math commands.

If you see the dollar signs and backslashes literally, your current Markdown viewer may not support math in that context.

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12. Five beginner mistakes

No space after a heading marker

```
##Results      wrong
## Results     better
```

No blank line between paragraphs

Use a blank line when you want a new paragraph.

Missing closing code fence

Every opening triple-backtick fence needs a closing fence.

Image path points to your own computer

Prefer:

```
![Result](figures/result.png)
```

instead of an absolute path from your desktop.

Assuming every Markdown viewer is identical

Basic syntax is very portable, but math, citations, footnotes, and some other features vary by tool.

For more fixes with examples, see `Markdown_Troubleshooting.pdf`.

Quick lookup

I want to…	Type
Heading	`# Heading`
Smaller heading	`## Heading`
Bold	`**text**`
Italic	`*text*`
Bullet	`- item`
Numbered item	`1. item`
Checkbox	`- [ ] task`
Link	`[name](https://...)`
Image	`![description](image.png)`
Inline code	`` `code` ``
Code block	three backticks before and after
Small table	cells separated by vertical bars, plus a separator row
Inline math	`$...$`
Display math	`$$...$$`

A complete first note

Copy this into a file called `paper_notes.md`:

```
# Paper Notes

**Paper:** Example Paper

## Main question

What question does the paper answer?

## Main finding

The main result is ...

## My questions

- Why is this sample used?
- What is the main mechanism?
- What should I check in the appendix?

## Tasks
```

```
- [ ] Read the data section
- [ ] Check Table 2
- [ ] Write a one-paragraph summary
```

Markdown is just as useful outside research. For example, a plan for the week:

```
# Plan for This Week

**Goal:** Finish the report.

## Monday

- [x] Outline the report
- [ ] Email the team

## Tuesday

- [ ] Write the first draft
```

That is already a useful Markdown document. Learn the advanced sheet only when you actually need more.